FMP4 reviewing feedback

INFOGRAPHIC | DATA VISUALIZATION | FOOD DATA | EDS 240

A data-driven exploration of how time, complexity, ingredients, and nutrition shape recipe popularity on Food.com.

AUTHOR
Joshua Ferrer-Lozano

PUBLISHED
March 8, 2026

Reviewing Feedback

1. Key feedback points I received on my FPM #3 draft visualizations Across peer review and instructor feedback, several themes emerged:

Clarity of annotations: Some labels overlapped or competed with the data, especially in the 30-minute plot and the complexity plot. Reviewers suggested repositioning annotations and improving hierarchy.

Axis readability: A few plots had axis labels or tick marks that were either too small or too dense.

Complexity plot labeling: Multiple reviewers suggested finding a more creative way to label the rating tiers instead of relying on a traditional x-axis.

Ingredient sunburst: Feedback encouraged simplifying ingredient categories and ensuring the hierarchy was intuitive.

Nutrition plot: Reviewers recommended choosing a single nutritional variable and/or faceting by category to avoid clutter and improve interpretability.

Overall cohesion: Several comments encouraged me to lean further into my editorial “menu” metaphor and unify the aesthetic across all plots (color, typography, spacing).

2. Which suggestions I implemented and how I incorporated nearly all major suggestions:

Improved annotation placement:
I repositioned labels in the 30-minute plot (e.g., moving the recipe count label downward, adjusting the 30-minute marker, and refining the curved arrow). This improved legibility and visual flow.

Refined axis readability:
I increased caption size, adjusted tick spacing, and used log-scaled review counts with intuitive labels (“1”, “10”, “100”, “1k”, “10k”).

Creative labeling for the complexity plot:
Instead of traditional x-axis labels, I added left-aligned text annotations beside each stacked bar. This created a cleaner, more editorial look and aligned with feedback to experiment with non-traditional labeling.

Ingredient sunburst simplification:
I filtered out generic ingredients (salt, pepper, water, oil, butter) and limited each category to its top five ingredients. This made the sunburst more interpretable and visually balanced.

Nutrition plot redesign:
Based on feedback, I selected sugar as the single nutritional variable and faceted the scatterplot by the three main categories (Dessert, Main Dish, Side Dish). This clarified the relationship between nutrition and popularity and avoided overplotting.

Cohesive aesthetic:
I standardized my palette (warm terracotta, caramel, cream), typography (Playfair Display + Lato), and background tone (#F5EFE6 / #F7F2EB). This created a unified editorial feel across all visualizations.

3. Feedback I chose not to implement and why Adding more categories to the nutrition facet:
   Some reviewers suggested including all recipe categories. I decided against this because it diluted the clarity of the comparison and made the layout visually overwhelming. Focusing on the three main categories preserved readability and narrative focus.

Adding more complexity tiers:
A suggestion was made to break complexity into more granular bins. I kept the three-tier system (Simple, Moderate, Complex) because it aligned better with my “layered cake” metaphor and maintained interpretability.

Using a different chart type for the 30-minute plot:
A few peers suggested a histogram or line plot. I kept the area chart because it visually reinforces the “mountain peak” of engagement at 30 minutes and fits the editorial tone.

4. Ten design elements I did not fully implement (and why) I addressed nearly all ten elements, but a few were less central to my project:

DEI lens:
The dataset (Food.com recipes) does not directly represent people or communities in a way that raises equity concerns. I still considered cultural categories carefully, but DEI was not a major design driver.

Accessibility through colorblind palettes:
My palette is warm and editorial, and while contrast is strong, it is not strictly colorblind-optimized. I prioritized aesthetic cohesion for this specific audience (general readers of a food-magazine-style infographic).

Contextualizing uncertainty:
Because my visualizations focus on descriptive patterns rather than inferential statistics, uncertainty intervals were not central to the story. I used LOESS smoothing where appropriate but did not emphasize statistical uncertainty.

Everything else — hierarchy, typography, color, annotation, spacing, narrative flow — was implemented intentionally and consistently.

What Makes a Recipe Popular? A Data-Driven Look at How We Cook at Home

Why do some recipes take off while others barely get noticed? It's a question every home cook, food blogger, and recipe platform has wondered about — and one that feels surprisingly personal. We all have dishes we return to again and again, and others we scroll past without a second thought. But what actually drives that difference at scale?

To explore this, I turned to a massive dataset of more than 188,000 recipes published on Food.com between 1999 and 2016. With rich metadata on ingredients, cooking times, ratings, and nutritional information, the dataset offered a rare opportunity to examine the patterns behind recipe engagement.

My goal was simple: understand what makes a recipe resonate with home cooks.

I approached this through three guiding questions:

- Time: Do quicker recipes attract more engagement?
- Complexity: Are simpler dishes more popular, or do cooks reward effort?
- Ingredients & Nutrition: What defines the "identity" of different recipe categories, and does nutrition influence popularity?

The result is the infographic below — a three-course exploration of how we cook, what we choose, and why certain recipes rise to the top.

Final Infographic

(Insert the PNG or rendered plot here — centered and sized appropriately.)

Designing the Infographic: A Behind-the-Scenes Look

Creating this infographic was as much a design challenge as a data challenge. Below, I walk through the ten design elements that shaped my process and how each contributed to the final product.

1. Hierarchy & Visual Flow

I structured the infographic like a meal: a starter, a main dish, a side, and a dessert.

This metaphor created a natural visual flow and helped me organize the story:

- The 30-minute plot opens the narrative with a clear, intuitive insight.
- The complexity "layered cake" plot anchors the center.
- The ingredient sunburst and sugar-popularity scatterplot act as side dishes that deepen the story.

This structure guides the viewer through the infographic the way a menu guides a diner.

2. Color

I used a warm, food-inspired palette — terracotta, caramel, cream, and sage — inspired by Bon Appétit and Sûr La Table. These tones:

- evoke a culinary aesthetic
- differentiate plot elements without overwhelming
- maintain cohesion across all visualizations

The palette also reinforces the editorial tone I wanted.

3. Typography

I paired Playfair Display for titles with Lato for body text and annotations. This serif-sans pairing is common in modern food media and strikes a balance between elegance and readability.

4. Annotation

Annotations were essential for turning raw data into a narrative:

- “30-Minute Sweet Spot” label highlights the dramatic drop-off in engagement.
- A curved arrow emphasizes the decline after 30 minutes.
- The complexity plot uses left-aligned labels instead of a traditional x-axis, creating a cleaner editorial look.
- The sugar-popularity plot includes intuitive log-scale labels (“1”, “10”, “100”, “1k”, “10k”).

These annotations help the viewer understand the story without needing to decode the data.

5. Layout

I used generous spacing, centered titles, and consistent margins to maintain an airy, magazine-style feel. Circular elements (like the sunburst and radial calorie plot) balance the rectangular bar charts and scatterplots, creating visual variety without clutter.

6. Simplicity

Each visualization focuses on a single idea:

- Time → engagement
- Complexity → rating tiers
- Ingredients → category identity
- Sugar → popularity

I intentionally avoided overplotting and limited categories where necessary (e.g., top five ingredients per category, three main facets for the sugar plot).

7. Consistency

All plots share:

- the same background tone (#F5EFE6 / #F7F2EB)
- the same font system
- similar annotation styling
- a warm, cohesive palette

This consistency makes the infographic feel like a unified piece rather than a collection of separate charts.

8. Accessibility

I used:

- high-contrast text
- clear labels
- intuitive axis scales
- non-color cues (stacking order, spacing)

While the palette is not strictly colorblind-optimized, contrast remains strong and readability is preserved.

9. Storytelling

Each visualization answers a sub-question, but together they reveal a broader pattern:

- Home cooks overwhelmingly prefer quick recipes.
- Moderate complexity is the sweet spot across rating tiers.
- Ingredient families define the identity of recipe categories.

- Nutrition — specifically sugar — plays a subtle but category-dependent role in popularity.

The narrative builds from practical (time) to structural (ingredients) to evaluative (ratings and nutrition).

10. Applying a DEI Lens

The dataset does not directly represent people or communities in a way that raises equity concerns, but I was mindful of how I categorized cuisines and avoided reinforcing stereotypes. I focused on descriptive patterns rather than value judgments.

Closing Thoughts

This project taught me how messy real-world food data can be — and how much care is required to clean, parse, and interpret it responsibly. From ISO 8601 time strings to inconsistent ingredient formatting, the preprocessing shaped what was possible in the final visualizations.

But the biggest takeaway is this: Popularity isn't random. It reflects how people actually cook.

Home cooks gravitate toward quick, moderately complex dishes built from familiar ingredients. The patterns are surprisingly consistent across categories and years.

In the end, this infographic is both a data story and a culinary one — a look at the rhythms, preferences, and habits that shape how we cook and eat at home.

The Data

Parse ISO 8601 Time

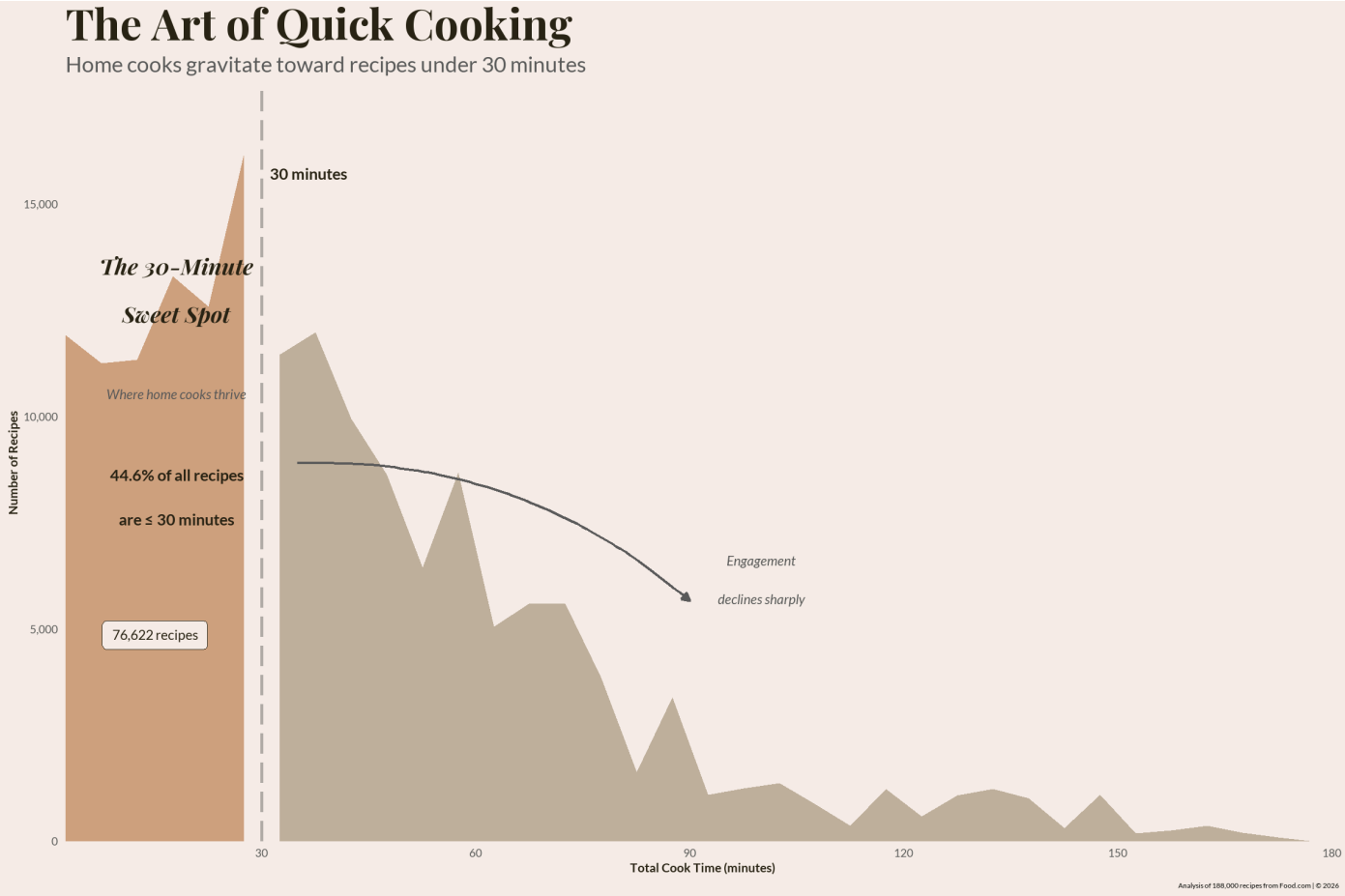
Apply Parsing

Fix Sugar Column

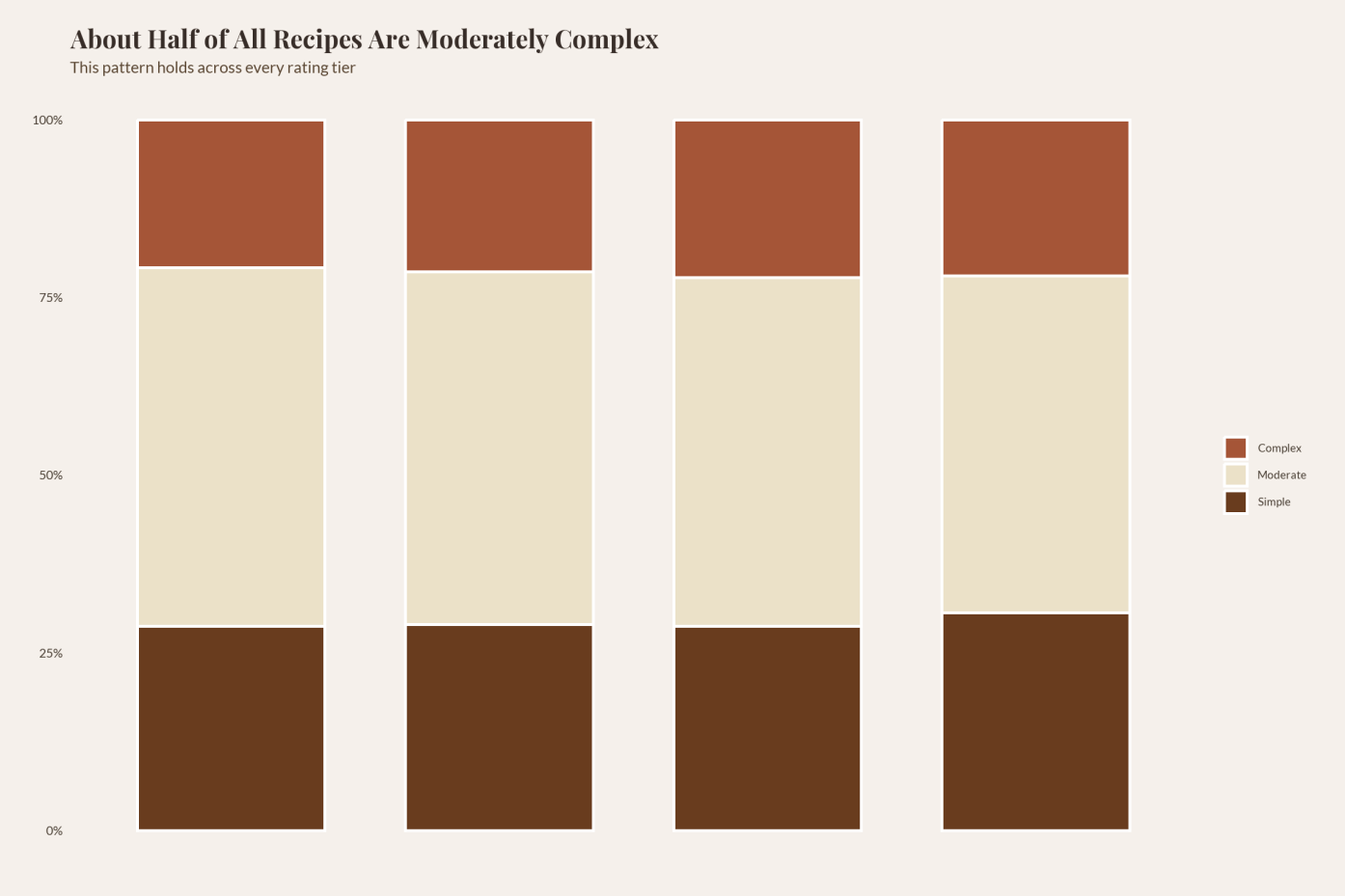
```
recipes_clean <- recipes_clean |>
  mutate(
    sugar_content = as.numeric(sugar_content)
  ) |>
  filter(
    sugar_content >= 0,
    sugar_content <= 500   # optional cap for sanity
  )

# recipes_clean <- recipes_clean |>
#   mutate(
```

```
#     sugar_content = readr::parse_number(sugar_content)
#   ) |>
#   filter(
#     sugar_content >= 0,
#     sugar_content <= 500
#   )
```

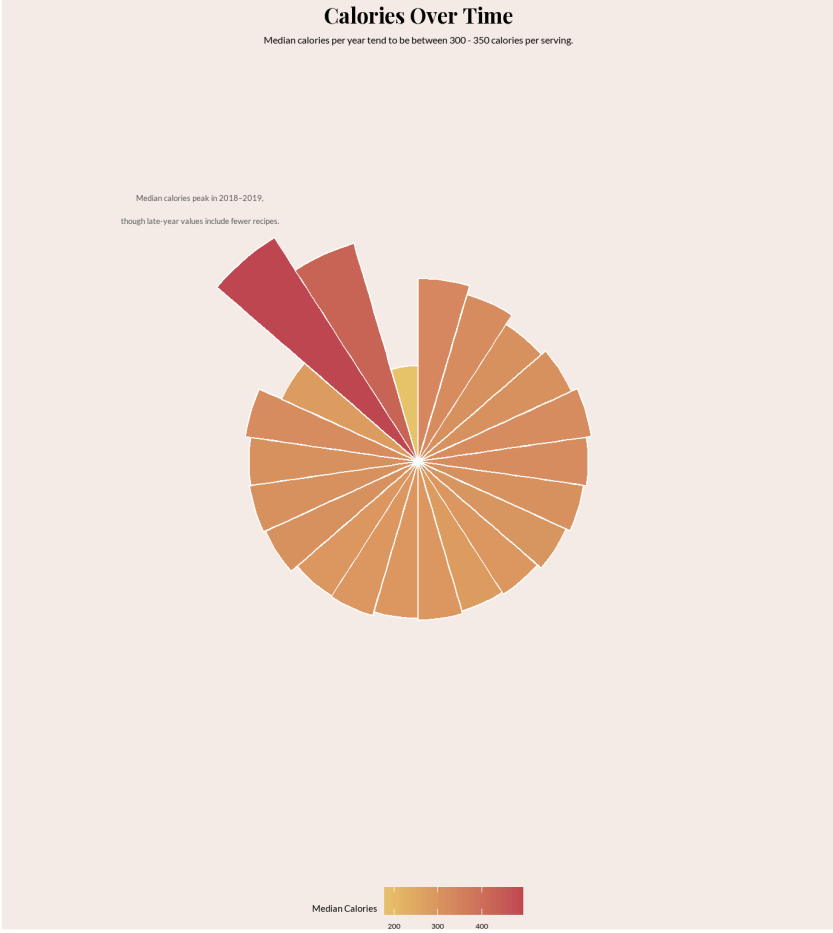
Visual 1: The 30-Minute Sweet Spot



Visual 2: Recipe Complexity by Rating Tier



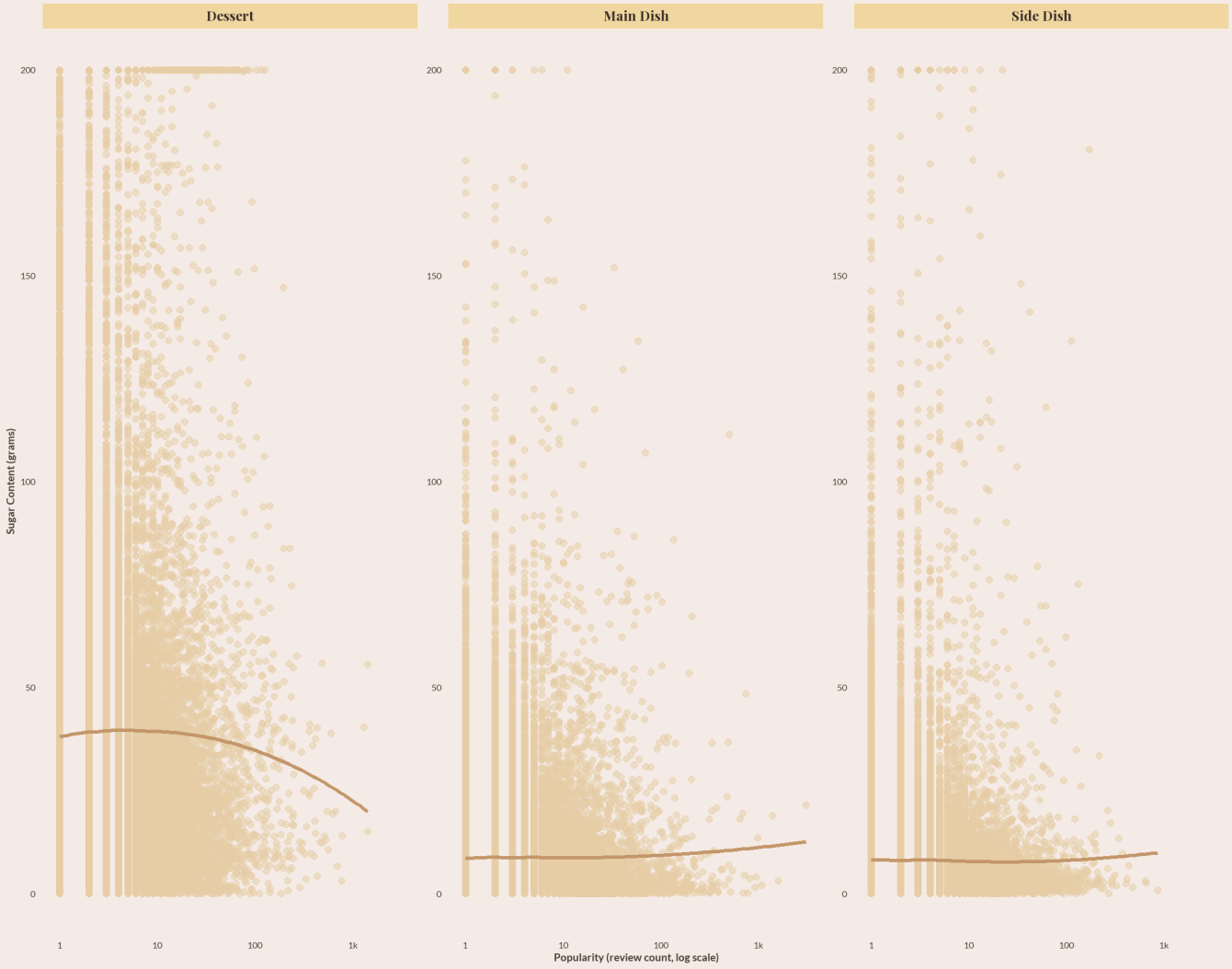
#Visual 3: Calories Over Time (Radial)

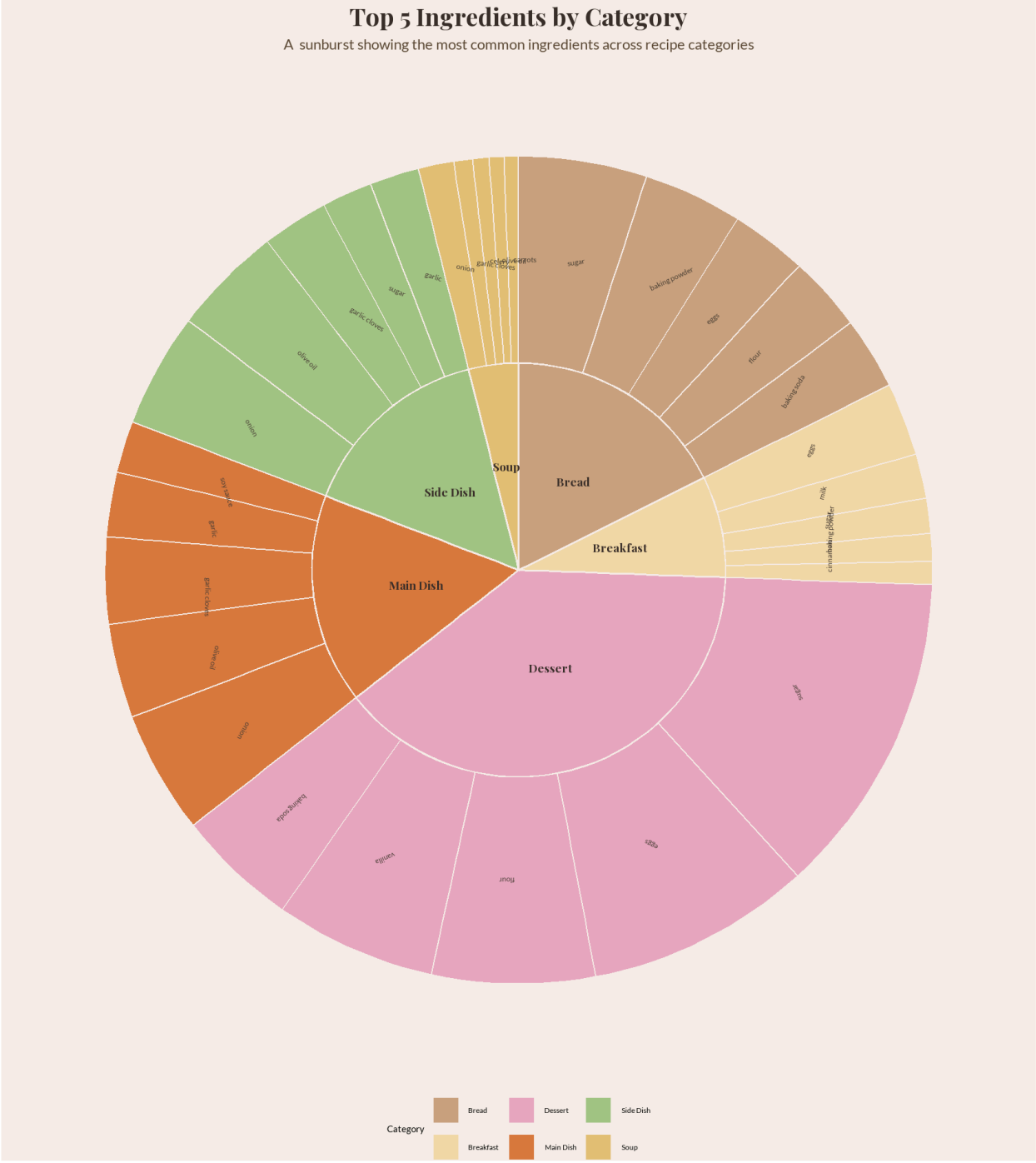

Calories Over Time
Median calories per year tend to be between 300 - 350 calories per serving.
Median calories peak in 2018–2019,
though late-year values include fewer recipes.
Median Calories
200
300
400

Sugar Plot

A Sweet Look at Popularity

Sugar content bubbles upward across recipes with different levels of engagement






```
# A tibble: 10 × 30
   name           author_name prep_time cook_time total_time date_published
   <chr>          <chr>       <chr>     <chr>     <chr>      <dttm>
 1 Biryani        elly9812    PT4H      PT25M     PT4H25M    1999-08-29 13:12:00
 2 Best Blackbot… Barefoot B… PT20M     PT2H      PT2H20M    1999-08-21 10:35:00
 3 Chai Tea       Linda Sund… PT15M     PT25M     PT40M      1999-08-16 05:10:00
 4 Cafe Cappucci… troyh       PT5M      <NA>      PT5M       1999-08-31 21:05:00
 5 Jimmy G's Car… Jimg2040    PT1H5M    PT45M     PT1H50M    1999-09-08 15:39:00
 6 Blueberry Des… Nancy Van … PT35M     <NA>      PT35M      1999-08-16 05:59:00
 7 Bourbon Pecan… benluc      PT1H      PT1H      PT2H       1999-09-01 09:01:00
 8 Chaat Masala   Carole Reu  PT25M     <NA>      PT25M      1999-08-26 19:39:00
 9 Carrot Ginger… Carole Reu  PT35M     PT15M     PT50M      1999-08-14 19:52:00
10 Catherine's E… Richard Ja… PT15M     PT20M     PT35M      1999-09-06 04:56:00
```

```
# ℹ 24 more variables: recipe_category <chr>, keywords <chr>,
#   recipe_ingredient_quantities <chr>, recipe_ingredient_parts <chr>,
#   aggregated_rating <dbl>, review_count <dbl>, calories <dbl>,
#   fat_content <dbl>, saturated_fat_content <dbl>, cholesterol_content <dbl>,
#   sodium_content <dbl>, carbohydrate_content <dbl>, fiber_content <dbl>,
#   sugar_content <dbl>, protein_content <dbl>, recipe_servings <dbl>,
#   category_group <chr>, year_published <dbl>, ingredient_count <dbl>, …
# A tibble: 45 × 30
# Groups:   category_group [9]
   name           author_name prep_time cook_time total_time date_published
   <chr>          <chr>       <chr>     <chr>     <chr>      <dttm>
 1 My Family's F… Hey Jude    PT15M     PT30M     PT45M      2003-06-05 20:03:00
 2 The Best Chil… AmandaAOat… PT30M     PT2H      PT2H30M    2003-10-10 20:00:00
 3 Roasted Green… Seamouse    PT5M      PT20M     PT25M      2005-08-17 15:06:00
 4 Easy and Tast… Kree6528    PT10M     PT8H      PT8H10M    2004-07-13 20:00:00
 5 Caramelized B… djmastermum PT5M      PT1H30M   PT1H35M    2005-11-14 17:12:00
 6 Ice Cubes      CHRISSYG    PT2M      PT6H      PT6H2M     2010-04-14 12:47:00
 7 Southern Swee… ratherbesw… PT15M     PT15M     PT30M      2003-06-05 20:09:00
 8 Iced Coffee    CoffeeMom   PT10M     PT3H      PT3H10M    2005-03-09 20:00:00
 9 Sangria        Bev I Am    PT35M     <NA>      PT35M      2002-11-16 20:11:00
10 Strawberry Ju… Heather Su… PT5M      <NA>      PT5M       2002-03-31 16:48:00
# ℹ 35 more rows
# ℹ 24 more variables: recipe_category <chr>, keywords <chr>,
#   recipe_ingredient_quantities <chr>, recipe_ingredient_parts <chr>,
#   aggregated_rating <dbl>, review_count <dbl>, calories <dbl>,
#   fat_content <dbl>, saturated_fat_content <dbl>, cholesterol_content <dbl>,
#   sodium_content <dbl>, carbohydrate_content <dbl>, fiber_content <dbl>,
#   sugar_content <dbl>, protein_content <dbl>, recipe_servings <dbl>, …

# A tibble: 45 × 30
# Groups:   category_group [9]
   name           author_name prep_time cook_time total_time date_published
   <chr>          <chr>       <chr>     <chr>     <chr>      <dttm>
 1 Cheese Rolls   Doreen Ran… PT45M     <NA>      PT45M      1999-09-01 12:34:00
 2 Grilled Avoca… agasdf      PT15M     PT15M     PT30M      1999-09-20 06:43:00
 3 Ham and Chees… Nancy Van … PT35M     <NA>      PT35M      1999-08-15 07:28:00
 4 Linda'a Guada… lindaWWJD   PT20M     <NA>      PT20M      1999-08-21 11:20:00
 5 Linda's Tuna … lindaWWJD   PT20M     PT1M      PT21M      1999-08-21 11:20:00
 6 Chai Tea       Linda Sund… PT15M     PT25M     PT40M      1999-08-16 05:10:00
 7 Cafe Cappucci… troyh       PT5M      <NA>      PT5M       1999-08-31 21:05:00
 8 Killer Kahlua  HELEN PEAG… PT35M     <NA>      PT35M      1999-09-15 03:38:00
 9 Cocoa Espress… Teddi2      PT20M     <NA>      PT20M      1999-08-30 13:58:00
10 Iced Jazzberr… Teddi2      PT30M     <NA>      PT30M      1999-08-30 13:58:00
# ℹ 35 more rows
# ℹ 24 more variables: recipe_category <chr>, keywords <chr>,
#   recipe_ingredient_quantities <chr>, recipe_ingredient_parts <chr>,
#   aggregated_rating <dbl>, review_count <dbl>, calories <dbl>,
#   fat_content <dbl>, saturated_fat_content <dbl>, cholesterol_content <dbl>,
#   sodium_content <dbl>, carbohydrate_content <dbl>, fiber_content <dbl>,
#   sugar_content <dbl>, protein_content <dbl>, recipe_servings <dbl>, …
```

```
# A tibble: 1 × 3
  min_reviews median_reviews max_reviews
        <dbl>          <dbl>       <dbl>
1           1              2        3063
```

Full Foldable Code

▶ Code